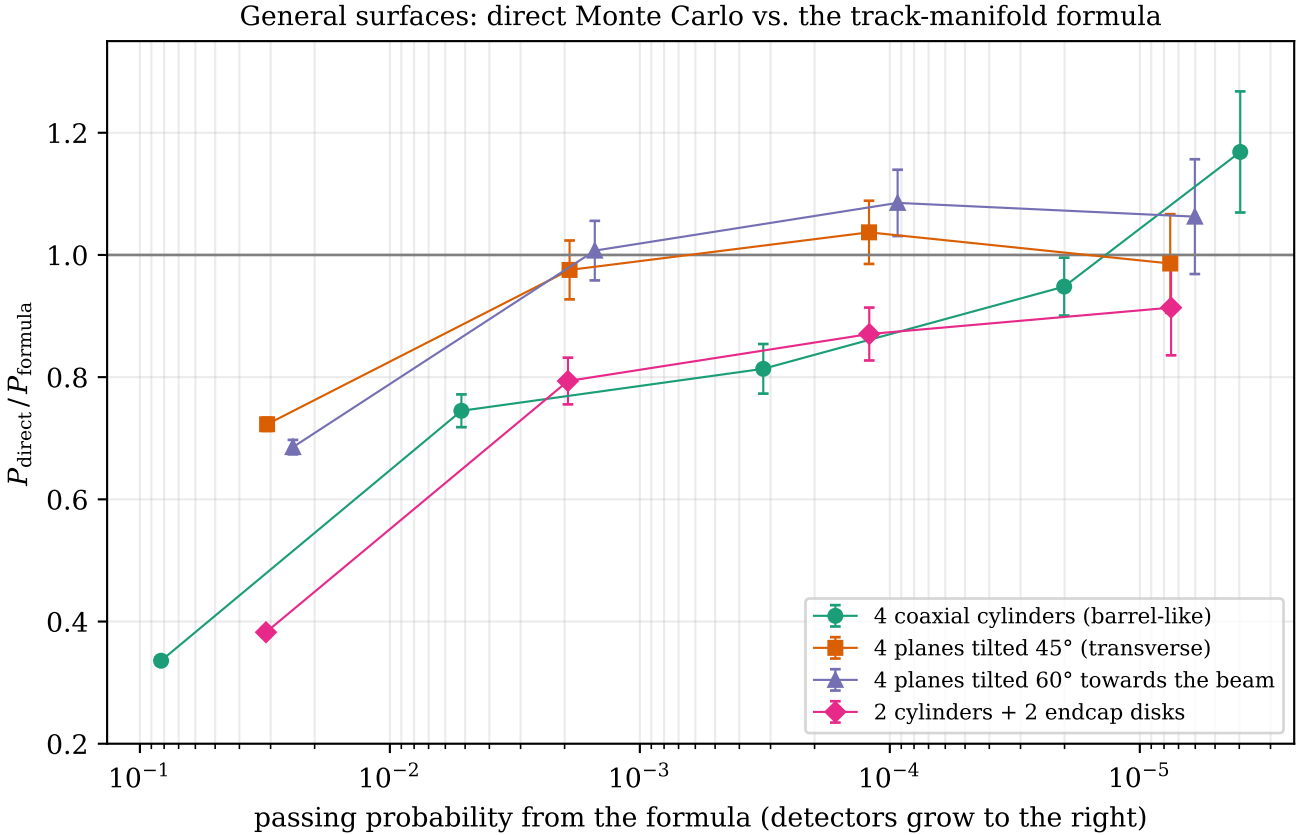
General surfaces: direct Monte Carlo vs. the track-manifold formula
$P_{\mathrm{direct}} / P_{\mathrm{formula}}$
1.2
1.0
0.8
0.6
0.4
0.2
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
passing probability from the formula (detectors grow to the right)
4 coaxial cylinders (barrel-like)
4 planes tilted 45° (transverse)
4 planes tilted 60° towards the beam
2 cylinders + 2 endcap disks